

Mechanism of free fatty acid-induced insulin resistance in humans.



To examine the mechanism by which lipids cause insulin resistance in humans, skeletal muscle glycogen and glucose-6-phosphate concentrations were measured every 15 min by simultaneous ^{13}C and ^{31}P nuclear magnetic resonance spectroscopy in nine healthy subjects in the presence of low (0.18 ± 0.02 mM [mean $\pm$ SEM]; control) or high (1.93 ± 0.04 mM; lipid infusion) plasma free fatty acid levels under euglycemic (approximately 5.2 mM) hyperinsulinemic (approximately 400 pM) clamp conditions for 6 h. During the initial 3.5 h of the clamp the rate of whole-body glucose uptake was not affected by lipid infusion, but it then decreased continuously to be approximately 46% of control values after 6 h ($P < 0.00001$). Augmented lipid oxidation was accompanied by a approximately 40% reduction of oxidative glucose metabolism starting during the third hour of lipid infusion ($P < 0.05$). Rates of muscle glycogen synthesis were similar during the first 3 h of lipid and control infusion, but thereafter decreased to approximately 50% of control values (4.0 ± 1.0 vs. 9.3 ± 1.6 $\mu\text{mol}/[\text{kg} \cdot \text{min}]$, $P < 0.05$). Reduction of muscle glycogen synthesis by elevated plasma free fatty acids was preceded by a fall of muscle glucose-6-phosphate concentrations starting at approximately 1.5 h (195 ± 25 vs. control: 237 ± 26 mM; $P < 0.01$). Therefore in contrast to the originally postulated mechanism in which free fatty acids were thought to inhibit insulin-stimulated glucose uptake in muscle through initial inhibition of pyruvate dehydrogenase these results demonstrate that free fatty acids induce insulin resistance in humans by initial inhibition of glucose transport/phosphorylation which is then followed by an approximately 50% reduction in both the rate of muscle glycogen synthesis and glucose oxidation.